Revision Tutorial 1

Microeconomics

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PART 1

Multiple-choice questions

Question 1(i)

Varian, Chapter 1

Pareto

"A change that makes at least one person better off without making anybody worse off". This is the definition of:

(a) An equitable reform
(b) A productive variation
(c) Pareto efficiency
(d) A Pareto improvement

Question 1(ii)

Varian, Chapter 3

Properties of preferences

Suppose that Barney's musical tastes are such that he prefers Jazz music to Rock music, Rock music to Classical music and Classical music to Jazz music. Barney's preferences do not display:

(a) Reflexivity
(b) Transitivity
(c) Completeness
(d) Both (a) and (b)
(e) All of (a), (b) and (c)

Question 1(iii)

Varian, Chapter 2

Budget line

Suppose there are two goods, ***x*** and ***y***, the prices of both goods are positive and a consumer's income is also positive. Suppose that the prices of good ***x*** and good ***y*** both double and income triples. On a graph where the budget line is drawn with the quantity of ***x*** on the horizontal axis and the quantity of ***y*** on the vertical axis,

(a) the budget line becomes steeper and shifts outward.
(b) the budget line becomes flatter and shifts outward.
(c) the new budget line is parallel to the old budget line and lies below it.
(d) the new budget line is parallel to the old budget line and lies above it.
(e) None of the above.

Question 1(iv)

Varian, Chapter 6

Types of goods

"A good for which demand increases as income falls." This is the definition of:

(a) a Normal good
(b) an Ordinary good
(c) an Inferior good
(d) a Giffen good

Question 1(v)

Varian, Chapters 6 and 8

Inferior and Giffen goods

Which of the following statements is correct?

(a) Giffen goods are always inferior goods
(b) Inferior goods are always Giffen goods
(c) Not all inferior goods are also Giffen
(d) (a) and (b) are both correct
(e) (a) and (c) are both correct

Question 1(vi)

Varian, Chapter 9

Labour supply

If leisure is a normal good, then an increase in income that does not involve a change in the wage rate (e.g. an increase in non-labour income) will tend to result in:

(a) An increase in labour supply
(b) A fall in labour supply
(c) No change in labour supply
(d) Any of (a), (b) or (c) are equally likely

PART 2

Cobb–Douglas demand

Exercise 2

Varian, Chapter 5

Tracey's choice

Tracey's utility function between consumption of heating fuel (bought in quantity f) and consumption of other goods (quantity g) may be described by

$$u(f,g) = f^{0.5}g^{1.5}$$

So marginal utilities are

$$MU_f = \frac{\partial u}{\partial f} = 0.5\,f^{-0.5}g^{1.5} \qquad \text{and} \qquad MU_g = \frac{\partial u}{\partial g} = 1.5\,f^{0.5}g^{0.5}$$

and her budget is a fixed money income m.

(a) Denote the price of fuel by P_f and the price of other goods by P_g. What is the equation for Tracey's budget constraint?

(b) Using the condition for optimality that relates the marginal rate of substitution and prices, derive an expression for the relationship between (optimally chosen) expenditure on other goods and (optimally chosen) expenditure on fuel.

Exercise 2 (continued)

Varian, Chapter 5

Tracey's choice

(c) Solve for the demand functions for ***f*** and ***g***.

(d) Suppose Tracey is observed at three points in time and the only things that change between points in time are Tracey's income level and prices. At the points in time, Tracey's expenditure on fuel and her total income are observed to be:

	Expenditure on fuel	Total income
Time 1	25	100
Time 2	30	120
Time 3	20	80

Are these expenditure patterns consistent with the utility function proposed in the question?

PART 3

Labour supply with a benefit and a tax

Exercise 3

Varian, Chapter 9

Labour–leisure choice

Suppose that an individual has 20 hours a day (!) available to allocate between work (***L***) or leisure (***R***). Suppose that he receives a wage rate of 10 dollars an hour, and that the other good that provides him with utility is consumption spending (***C***), measured in dollars.

(a) Draw the budget line for this consumer's problem, putting ***R*** on the horizontal axis and being careful to label axes and the intercepts of the curve.

Exercise 3 (continued)

Varian, Chapter 9

A benefit and a tax

Now suppose that the government introduces a benefit payment that boosts income by 20 dollars per individual per day. To pay for this benefit, the government introduces a tax which takes 20 cents of each dollar of earned income and which starts being paid once labour income (i.e. income excluding the 20 dollars of benefit) exceeds 20 dollars per day.

(b) At what number of hours of work will the tax start to apply for the individual considered in this question?

(c) Including the effects of the benefit payment and of the tax, what will be the individual's income if he works: (i) 0 hours; (ii) 5 hours; (iii) 10 hours; (iv) 20 hours.

(d) On your graph for part (a), draw a new budget line that represents the case with the benefit and the tax both affecting income. Be careful to label axes, the intercepts of the curve, and the point at which the tax starts to apply.